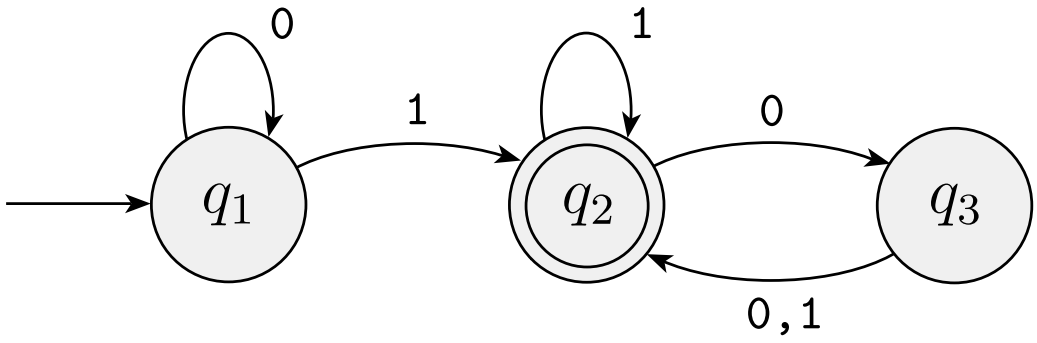
Nondeterministic Finite Automata

Alessandro Chiesa

EPFL School of Computer and Communication Sciences

Lecture 2



RECALL: DFAs, REGULAR LANGUAGES

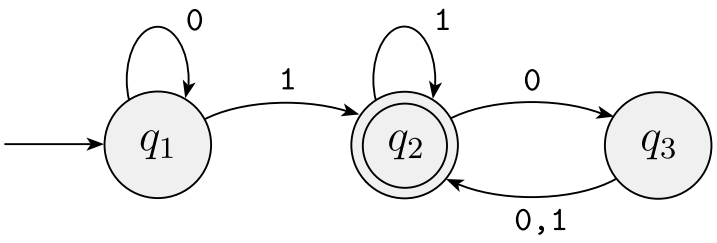
Definitions

A deterministic finite automaton (DFA) M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

- Q is a finite set called the states,
- Σ is a finite set called the alphabet,
- $\delta : Q \times \Sigma \to Q$ is the transition function,
- $q_0 \in Q$ is the start state, and
- $F \subseteq Q$ is the set of accept states. *(allow $F = \emptyset$)*

- $Q = \{q_1, q_2, q_3\}$,
- $\Sigma = \{0, 1\}$,
- δ is described as

	0	1
q_1	q_1	q_2
q_2	q_3	q_2
q_3	q_2	q_2,



- q_1 is the start state, and
- $F = \{q_2\}$.

Definitions

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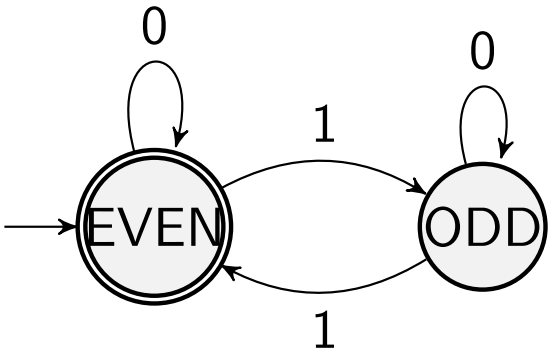
- Σ^* — set of all finite strings composed of symbols from Σ
  - Includes the empty string ε
- $L(M) \subseteq \Sigma^*$ set of strings accepted by M. We say that M recognizes the language $L(M)$.
- $A \subseteq \Sigma^*$ is a regular language iff there is a DFA M s.t. $A = L(M)$.

Is $L = \{w \in \{0,1\}^* : w \text{ contains an even number of 1's}\}$ a regular language?

YES because it is recognized by the following DFA



Formally, we prove correctness, i.e., that a DFA accepts a certain language, via induction on the length of the string. It is a good idea to choose a strong induction hypothesis that classifies which strings leads to which states instead of only talking about accepting states

NEW LANGUAGES FROM OLD

- Complement
  - $\bar{L} = \{w \in \Sigma^* : w \text{ is } \textbf{not} \text{ in } L\}$
- Union
  - $L_1 \cup L_2 = \{w \in \Sigma^* : w \in L_1 \textbf{ or } w \in L_2\}$
- Intersection
  - $L_1 \cap L_2 = \{w \in \Sigma^* : w \in L_1 \textbf{ and } w \in L_2\}$
- Concatenation
  - $L_1 \circ L_2 = \{w \in \Sigma^* : w = w_1.w_2,\ w_1 \in L_1 \textbf{ and } w_2 \in L_2\}$

Complement

If L is regular is its complement $\bar{L} = \{w \in \Sigma^* : w \text{ is } \textbf{not} \text{ in } L\}$ regular?

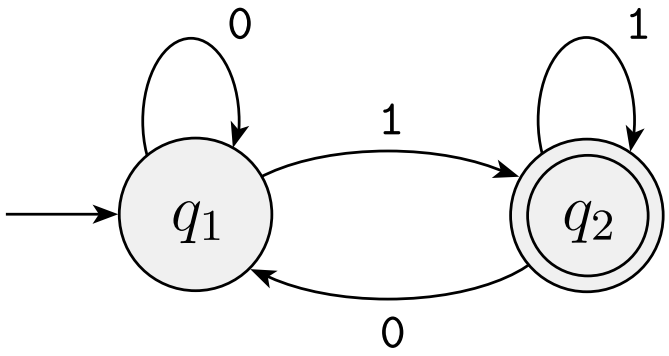
- Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA that accepts L
- Let $M' = (Q, \Sigma, \delta, q_0, \bar{F} = Q \setminus F)$ be the DFA where accepting states are complemented

- If $w \in L(M)$ then $w \notin L(M')$
- If $w \notin L(M)$ then $w \in L(M')$

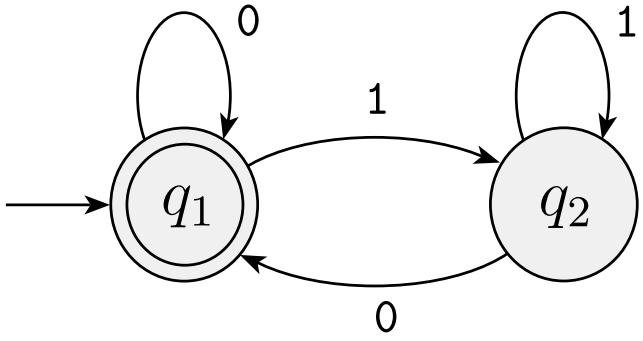
Theorem: $L(M') = \bar{L}$

Hence complement of a regular language is regular

Example



is the complement of



Union

- $L_1 = L(M_1), M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$
- $L_2 = L(M_2), M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$

If M_1's alphabet Σ_1 is different from M_2's alphabet Σ_2 then first extend them to the alphabet $\Sigma = \Sigma_1 \cup \Sigma_2$ before taking the union

The union is recognized by the DFA $M = (Q, \Sigma, \delta, q_0, F)$ where

- $Q = Q_1 \times Q_2$
- $\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$
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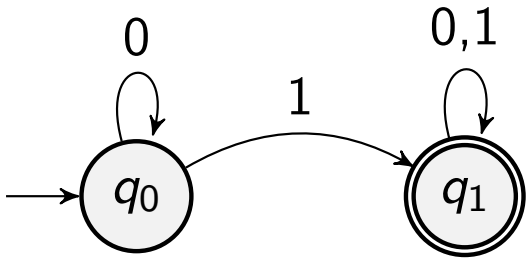
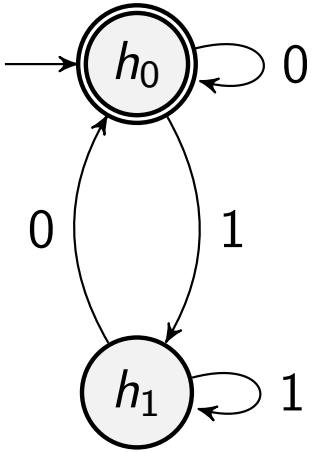
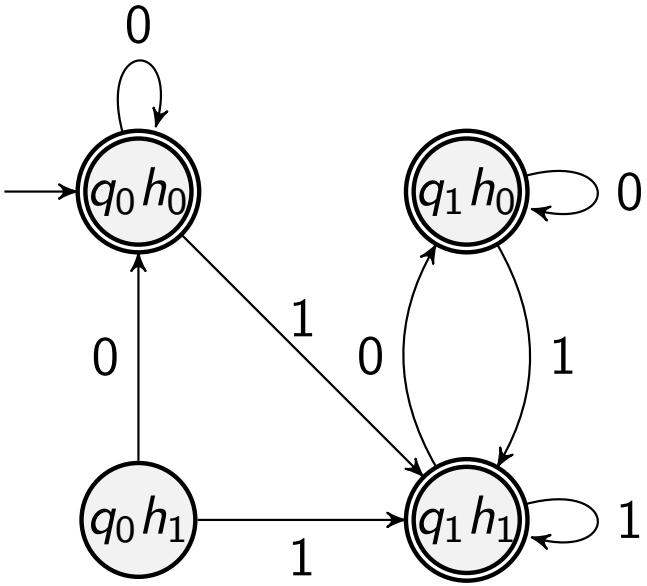
Run M_1 and M_2 in parallel and accept if one of them does

Example







The union

Intersection

- $L_1 = L(M_1), M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$
- $L_2 = L(M_2), M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$

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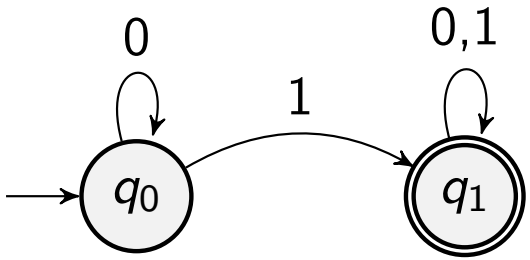
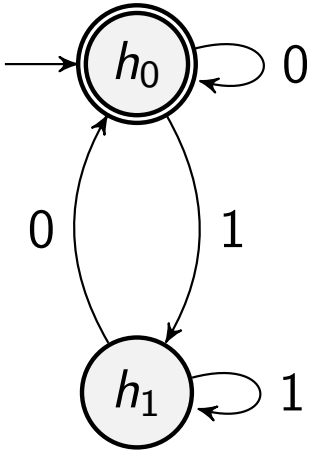
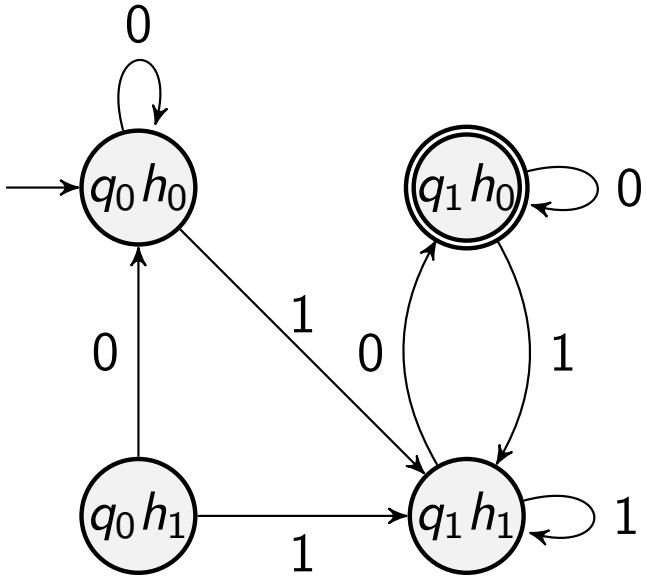
Example





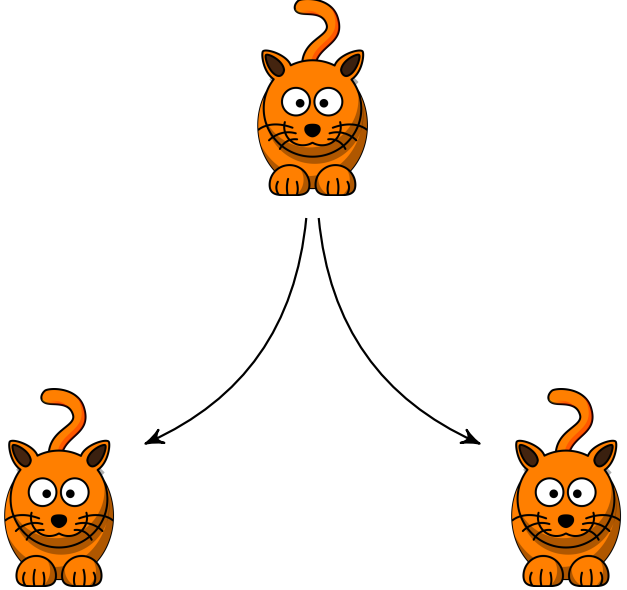


The intersection

Concatenation?

$$L_1 \circ L_2 = \{w \in \Sigma^* : w = w_1.w_2,\ w_1 \in L_1 \textbf{ and } w_2 \in L_2\}$$

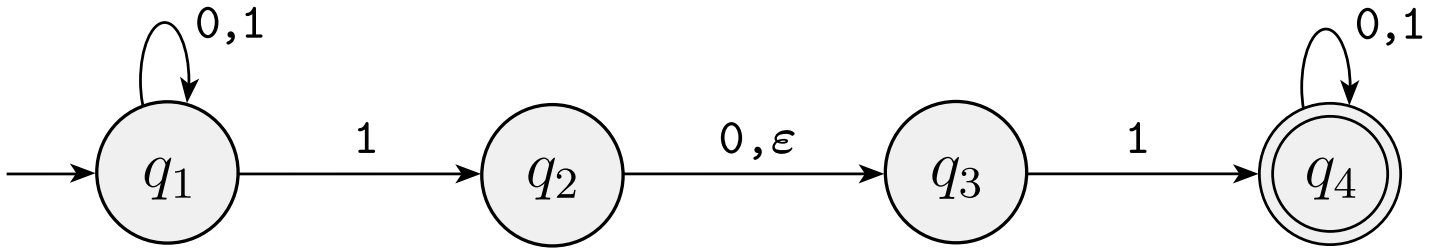
Difficulty to decide whether a string w is in $L_1 \circ L_2$ is that we need to "guess" where to break it into w_1 and w_2



Should I stay or should I go (left or right)

NONDETERMINISM

Nondeterminism vs Determinism



State diagram of a **Nondeterministic Finite Automaton (NFA)**

Differences to DFAs:

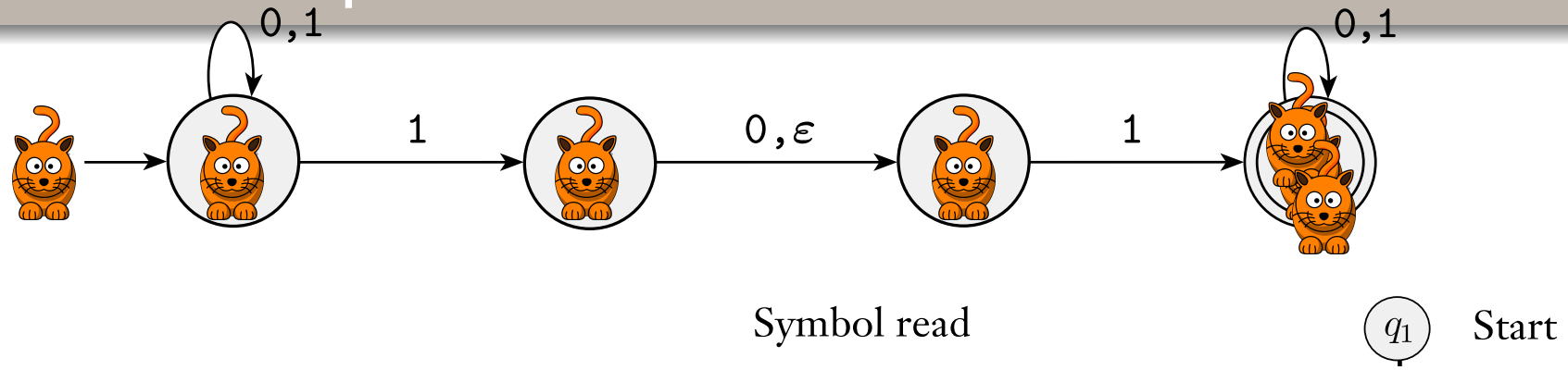
1. Ability to transition to more than one state on a given symbol
   Ex: q_1 has two outgoing transitions for symbol 1
2. A state may have no transition on a particular symbol
   Ex: q_3 has no outgoing transitions for symbol 0
3. Ability to take a step without reading any input symbol (ε-transitions)
   Ex: q_2 can transition to q_3 without reading a symbol

How does an NFA Compute?

Two different but equivalent viewpoints:

- It is a parallel computer that makes multiple copies of itself and follows *all* possibilities in parallel. If *any one* of these copies is in an accept state at the end, the NFA accepts the input string
- It is a computer that accepts if there exist guesses/choices that makes it accept

Parallel Viewpoint



Input: 010110
▲▲▲▲▲▲

Let's start

- cat moves to initial state q_1

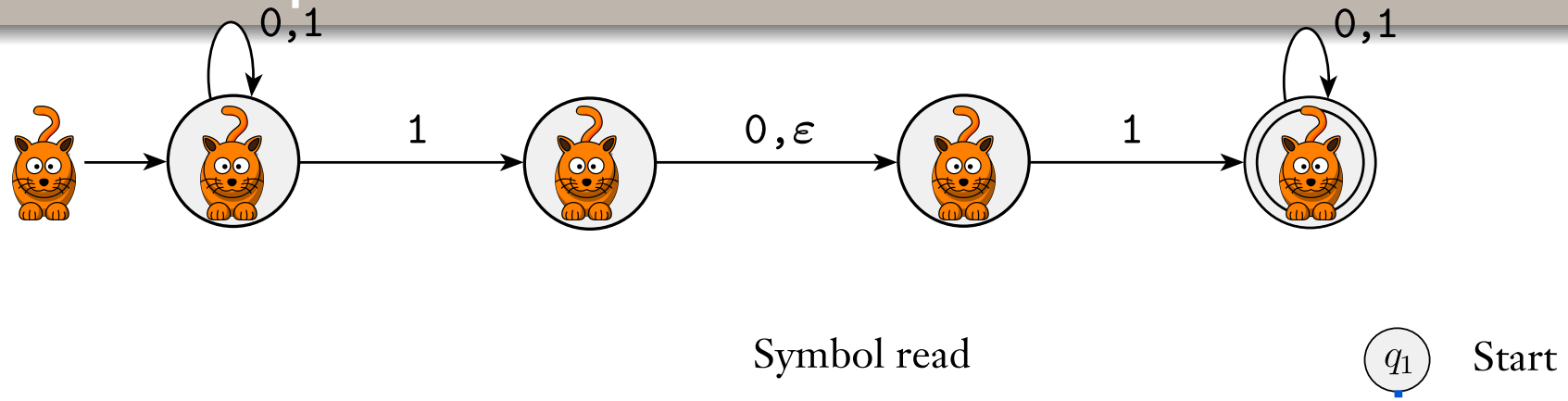
Symbol read 0

- cat stays in state q_1

Symbol read 1

- cat can (i) stay in state q_1, (ii) move to q_2, and (iii) move to q_2 and then to q_3 via the ε-transition
- We make a copy of the process (cat) for each of the three possibilities

Guess Viewpoint



Input: 010110

Let's start

- cat moves to initial state q_1

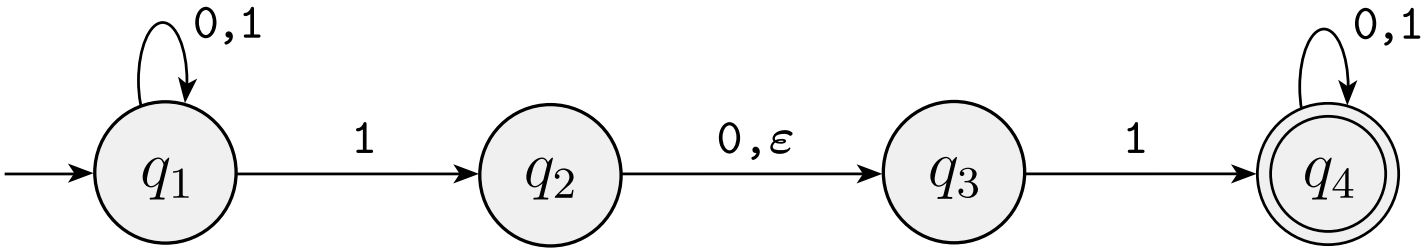
Symbol read 0

- cat stays in state q_1

Symbol read 1

- cat can (i) stay in state q_1, (ii) move to q_2, and (iii) move to q_2 and then to q_3 via the ε-transition
- we make a guess that the cat goes to state q_2

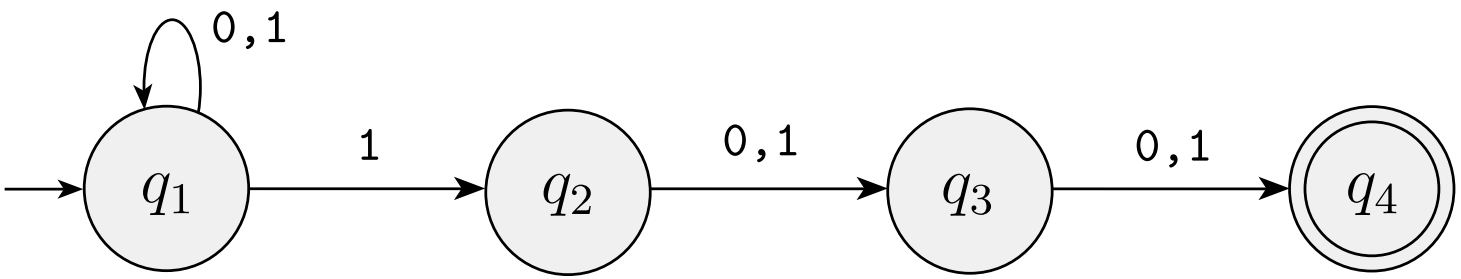
Example 1



What language does the NFA recognize?

$$L = \{w \in \{0,1\}^* \mid w \text{ contains 11 or 101 as a substring}\}$$

Example 2

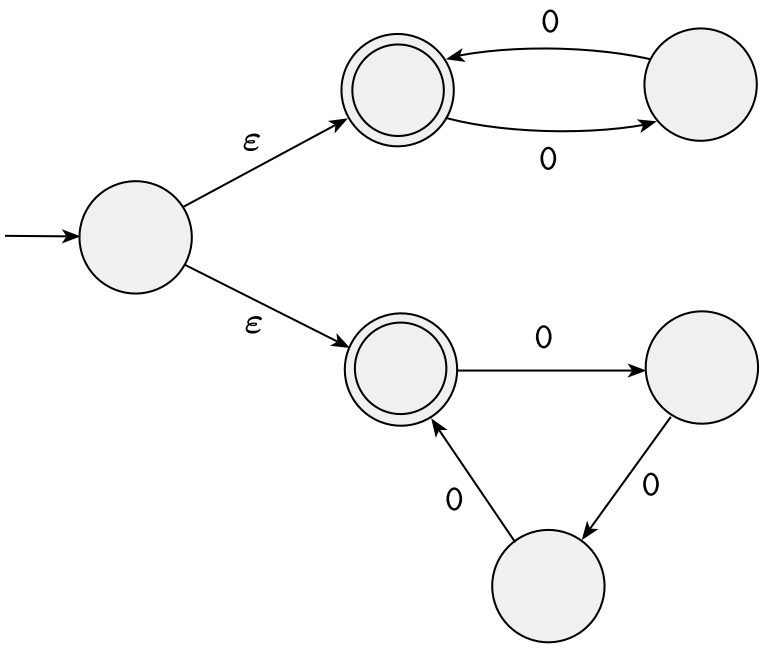


What language does the NFA recognize?

$$L = \{w \in \{0,1\}^* \mid w \text{ contains a 1 in the third position from the end}\}$$

Remark: If in the computation path you visit an accept state but don't end there, then that path is not accepting. E.g. 1000

Example 3



What language does the NFA recognize?

$$L = \{w \in \{0,1\}^* \mid w = 0^k \text{ where } k \text{ is divisible by 2 or 3}\}$$

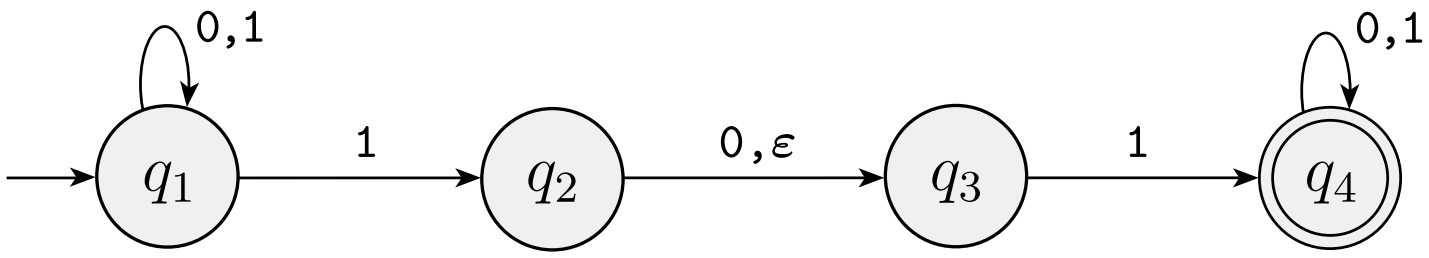
Formal definitions

A nondeterministic finite automaton (NFA) M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

- Q is a finite set called the states,
- Σ is a finite set called the alphabet,
- $\delta : Q \times (\Sigma \cup \{\varepsilon\}) \to 2^Q$ is the transition function,
- $q_0 \in Q$ is the start state, and
- $F \subseteq Q$ is the set of accept states. *(allow $F = \emptyset$)*

- Here 2^Q denotes the power set of Q. Ex: $Q = \{q_1, q_2\}, 2^Q = \{\emptyset, \{q_1\}, \{q_2\}, \{q_1, q_2\}\}$
- An input is accepted if there exists at least one path that ends at an accepting state
- An input is rejected if no computation path end at an accepting state

Example



The formal description of N_1 is $(Q, \Sigma, \delta, q_1, F)$, where

1. $Q = \{q_1, q_2, q_3, q_4\}$,

Why Study Nondeterminism?

If there is no "real-world" analog of nondeterminism construct, namely, make the choice that will lead to success, why bother?

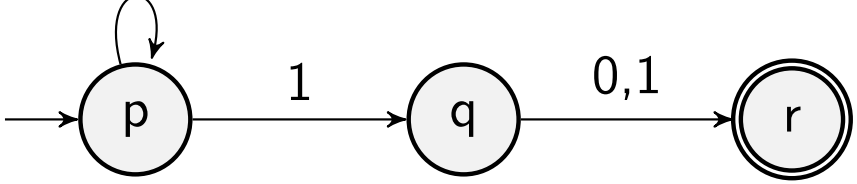
- Mathematically well-defined and valid question to study its impact on computational power
- Useful in translation of regular languages:
  - As we will show, every NFA can be converted into an equivalent DFA
  - So a language is regular iff it is recognized by an NFA
  - Constructing NFAs is sometimes easier than directly constructing DFAs (doing concatenation with NFA is an exercise)
  - NFA may be much (exponentially) smaller than its deterministic counterpart, or its functioning may be easier to understand
- Nondeterminism is also critical to theory of NP-complete problems which includes a large number of natural optimization problems

Theorem. Every nondeterministic finite automaton has an equivalent deterministic finite automaton.

Corollary. A language is regular if and only if some nondeterministic finite automaton recognizes it.

- We prove the theorem for NFAs without ε-transitions
- Exercises
  - Change the proof to work in the presence of ε-transitions (see book)
  - Given an NFA N construct an NFA N' with $L(N') = L(N)$ and no ε-transitions in N' (see Exercise Set 2)

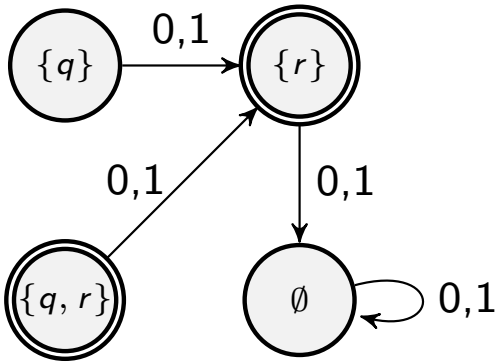
Example

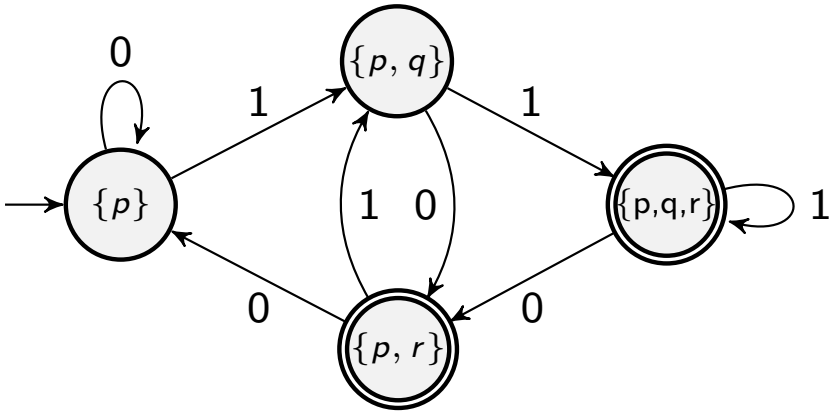


Define $\tilde{\delta} : 2^Q \times \Sigma \to 2^Q$ as the function that maps possible states $A \subseteq Q$ to the possible states after reading one more symbol $a \in \Sigma$:

$$\tilde{\delta}(A, a) = \bigcup_{q \in A} \delta(q, a)$$

$\tilde{\delta}$	**0**	**1**
$\emptyset$	$\emptyset$	$\emptyset$
$\{p\}$	$\{p\}$	$\{p, q\}$
$\{q\}$	$\{r\}$	$\{r\}$
$\{r\}$	$\emptyset$	$\emptyset$
$\{p, q\}$	$\{p, r\}$	$\{p, q, r\}$
$\{p, r\}$	$\{p\}$	$\{p, q\}$
$\{q, r\}$	$\{r\}$	$\{r\}$
$\{p, q, r\}$	$\{p, r\}$	$\{p, q, r\}$





Extended Transition Function

For a DFA: A state and a sequence of symbols maps to a state by sequentially applying transit function, by reading the input sequence letter by letter

$$M = (Q_M, \Sigma, \delta_M, q_0, F_M)$$

$$\Delta_M : Q_M \times \Sigma^* \to Q_M$$

$$\Delta_M(A, \varepsilon) = A$$
$$\Delta_M(A, xa) = \delta_M(\Delta_M(A, x), a)$$

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For an NFA: A set of states and a sequence of symbols maps to a set of states by sequentially applying transit function, by reading the input sequence letter by letter. A state with null input maps to the same state (no ε-transitions)

$$N = (Q_N, \Sigma, \delta_N, q_0, F_N) \qquad \Delta_N : 2^{Q_N} \times \Sigma^* \to 2^{Q_N}$$

$$\Delta_N(A, \varepsilon) = A$$
$$\Delta_N(A, xa) = \cup_{q \in \Delta_N(A,x)} \delta_N(q, a)$$

Extended Transition Function

For a DFA: A state and a sequence of symbols maps to a state by sequentially applying transit function, by reading the input sequence letter by letter

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$$\Delta_N(A, \varepsilon) = A$$
$$\Delta_N(A, xa) = \cup_{q \in \Delta_N(A,x)} \delta_N(q, a)$$

Lemma: For all $x, y \in \Sigma^*$, $\Delta_N(A, xy) = \Delta_N(\Delta_N(A, x), y)$

- Proof via induction on $|xy|$ — exercise

Theorem. Every nondeterministic finite automaton has an equivalent deterministic finite automaton

Proof

Take any NFA:

$$N = (Q_N, \Sigma, \delta_N, q_0, F_N)$$

Extended transition function:

$$\Delta_N : 2^{Q_N} \times \Sigma^* \to 2^{Q_N}$$
$$\Delta_N(A, \varepsilon) = A$$
$$\Delta_N(A, xa) = \cup_{q \in \Delta_N(A,x)} \delta_N(q, a)$$

We need to construct a DFA M that accepts the same language

$$M = (Q_M, \Sigma, \delta_M, q_0, F_M)$$

The Subset Construction:

$$Q_M = 2^{Q_N}$$
$$\delta_M(A, a) = \Delta_N(A, a)$$
$$F_M = \{A \subseteq Q_N : A \cap F_N \neq \emptyset\}$$

Proof of Theorem (continued)

Lemma. For all $A \subseteq Q_N, x \in \Sigma^*$, $\Delta_M(A, x) = \Delta_N(A, x)$

Proof of Lemma: Induction on |x|

Base case: $x = \varepsilon$

$$\Delta_M(A, \varepsilon) = A = \Delta_N(A, \varepsilon)$$

Induction Step: Let $x = ya$ where $a \in \Sigma$ and $a \neq \emptyset$

$$\begin{aligned}
\Delta_M(A, ya) &= \delta_M(\Delta_M(A, y), a) && \text{(Definition of } \Delta_M\text{)} \\
&= \delta_M(\Delta_N(A, y), a) && \text{(Induction as } |y| < |x|\text{)} \\
&= \Delta_N(\Delta_N(A, y), a) && \text{(By subset construction)} \\
&= \Delta_N(A, ya) && \text{(Definition of } \Delta_N\text{)}
\end{aligned}$$

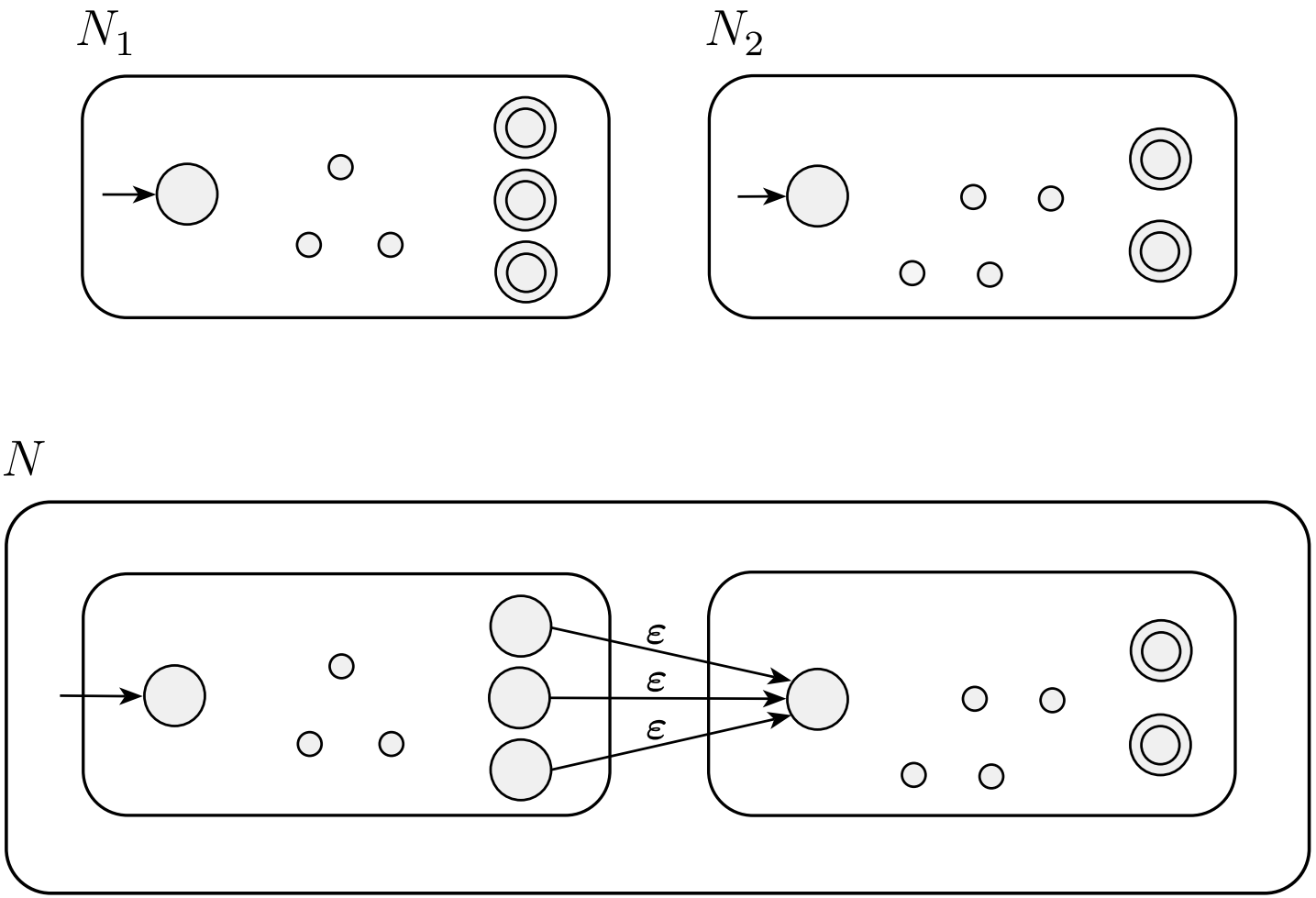
Lemma implies Theorem:

$$x \in L(M) \Leftrightarrow \Delta_M(q_0, x) \in F_M \Leftrightarrow \Delta_M(q_0, x) \cap F_N \neq \emptyset$$
$$\underbrace{\Longleftrightarrow}_{\text{Lemma}} \Delta_N(q_0, x) \cap F_N \neq \emptyset \Leftrightarrow x \in L(N)$$

Back to Concatenation

$$L_1 \circ L_2 = \{w \in \Sigma^* : w = w_1.w_2,\ w_1 \in L_1 \textbf{ and } w_2 \in L_2\}$$

Concatenation



Formal construction

Let $N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1,
and $N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2

Construct $N = (Q, \Sigma, \delta, q_1, F_2)$ to recognize $A_1 \circ A_2$

1. $Q = Q_1 \cup Q_2$ The states of N are all the states of N_1 and N_2
2. The state q_1 is the same as the start state of N_1
3. The accept states F_2 are the same as the accept states of N_2
4. Define δ so that for any $q \in Q$ and any $a \in \Sigma \cup \{\varepsilon\}$,

$$\delta(q, a) = \begin{cases} \delta_1(q, a) & q \in Q_1 \text{ and } q \notin F_1 \\ \delta_1(q, a) & q \in F_1 \text{ and } a \neq \varepsilon \\ \delta_1(q, a) \cup \{q_2\} & q \in F_1 \text{ and } a = \varepsilon \\ \delta_2(q, a) & q \in Q_2 \end{cases}$$

Is every Language Regular?